

CO₂LLAB: Creating an Eco-Conscious Community through Habit Tracking and Augmented Reality Visualisation

Hoi Yau Rosalyn Chun
University College London
hoi.chun.21@ucl.ac.uk

Yunhan Gao
University College London
ucjuyg0@ucl.ac.uk

Rahma Khairunisa Nursalamah
University College London
rahma.nursalamah.22@ucl.ac.uk

Conor Michael O'keeffe
University College London
ucjuco4@ucl.ac.uk

Haeji Shin
University College London
haeji.shin.22@ucl.ac.uk

ABSTRACT

CO₂e emissions remain a substantial problem for the environment despite efforts from numerous initiatives over the years to curb their output. The role of younger generations is critical in this problem space. From user research, we found that young adults rarely volunteer but do engage frequently in individual actions geared towards the environment. They are also motivated when they see others' contributions. We decided to leverage this willingness for individual environmental actions and turn it into a collective effort by creating CO₂LLAB, a platform that reimagines how young adults can reduce CO₂e emissions with their community. It incorporates a habit-tracking app for eco-friendly actions and augmented reality technology to visualise their impact. Our prototype evaluation demonstrated that CO₂LLAB not only educated them on the topic, but also motivated them to consider their impact on their community more.

KEYWORDS

Augmented Reality, Climate Action, CO₂e Emissions, Habit Tracker, Sustainable Community

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1 INTRODUCTION

Emissions of carbon dioxide, among other greenhouse gases, are the primary driver of climate change, one of the greatest problems facing humanity today. Within this remit, there is a well-established link between increases in temperature and carbon dioxide emissions [1]. In the UK alone, all of the top 10 warmest years since 1884 have occurred this century [2]. Man-made climate change has also been

cited as the main factor contributing to California's fivefold increase in the area of land burned by wildfires between 1972–2018 [3].

While it is accurate that most of these carbon emissions are produced by industry, individuals have a part to play in this problem too, and we must become more accountable to face the challenge. Take something so simple as ordering a coffee during your morning commute and having your drink in a disposable cup. Considering that an estimated 250 billion to 300 billion disposable cups are used annually worldwide [4], if individuals changed to a reusable cup, it could make a dramatic difference. Individual actions are a highly effective method to combat climate change, not to mention much swifter and easily enacted than government or industry-level efforts [5, 6]. However, individuals today have the tendency to underestimate the impact of their actions and show poor accuracy when calculating greenhouse gas emissions [7].

In order to reach the targets set out by the Paris Climate Agreement, carbon emissions need to be cut by 45% by 2030 and net zero by 2050 [8]. Accomplishing these ambitious targets will require educating people on how their seemingly innocent daily habits affect the environment and how the right changes to these can make a big impact. Meanwhile, communities also need to come together as a collective to make each other accountable for what needs to be done. As such, our project targets goals 13 and 11 of the UN Sustainable Development Goals, addressing climate action and fostering sustainable communities to drive this change.

2 USER GROUP

Our user group focus was young adults, specifically current university students. We chose this group as they will be responsible for driving social change regarding the UN Sustainable Development Goals in the future [9]. Initially, the scope of our research was quite broad as we tried to understand this generation's attitude towards social issues and their behaviours around contributions. We hypothesised from our preliminary research that engagement with charities – the main driver of social change in communities for previous generations – was lower among younger generations [10].

3 USER RESEARCH

An online survey with 48 young adults was carried out to gain a general understanding of how they contribute to social causes. Questions were about their attitudes towards charities, volunteer work, and individual actions as methods to contribute to social

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causes. An in-depth semi-structured interview was conducted afterwards to put these results into context and probe deeper. We chose 3 survey participants whose responses reflected different behaviour patterns of contributing to a social cause of interest. Some questions asked were:

How they decide which social cause to contribute
 If they share their contributions with others, how and why
 What they think about contributing to a social cause offline versus online
 What discourages people from contributing to the cause they care about

We also carried out semi-structured interviews with 7 charity organisations to understand their perspective towards young adults' engagement by asking questions such as:

Some issues they are facing as an association
 What the engagement of young adults are like
 How they engage young adults to participate

3.1 Findings: Young Adults

Our research revealed that it is not common for young adults to volunteer with charities frequently, with most participants either never volunteering or doing it on a yearly basis. Instead, individual actions, mostly directed towards the environment, such as using a reusable coffee cup and recycling, are common among young adults. This is mainly due to their efficiency in time and money, and because they alone are responsible for the action and are not reliant on others to perform it. People are open to exploring new methods of contributing and educating themselves about causes they weren't interested in before. However, they are somewhat siloed in their individual efforts as while most reported being motivated by seeing others' contributions, only a fraction felt comfortable posting about their cause online. One interviewee described it as "humble bragging" to post online about social contributions and another referred to it as a "mood breaker" for conversations, even offline. Individuals can also sometimes lose the motivation to contribute due to this lack of community, goals, and clear visibility of their impact.

Overall, young adults feel strongly about social causes and believe they are not doing enough with their current contributions. Trust in charities is generally an issue and validating the credibility of charities can also often take time, which is a precious resource for young people with their busy lifestyles. They also believe that charities could do a better job of connecting with them by providing more compelling and engaging content and relaying more feedback about the impact of their contributions.

3.2 Findings: Charities

Interviews with charities supported our survey findings that they are having trouble with recruiting young adults to volunteer by in large. From their perspective, young adults have very fluid lives with very busy schedules, which makes it difficult to get their time and attention. They also described how it was key to provide transparent information on the tangible impact that people's contributions make to motivate them to continue contributing and foster trust. One charity noted that they were more interested in creating a community of people who were intrinsically motivated

to contribute to them on a regular basis rather than receiving help on a more infrequent and transactional level.

As we analysed participants' feedback, we began to question whether the age-old charity model was suitable for younger generations when taking their needs, goals, and lifestyles into account.

3.3 User Needs

From our research findings, we developed two personas representing the user group (Figure 1). We also devised the user needs as being 1) to be engaged in contributing to social causes in new and novel ways, 2) to contribute together with socially conscious individuals, and 3) to see the impact that each person in the community makes.

4 DESIGN PROCESS

At this stage, we explored how young adults' goodwill and effort towards individual actions could be leveraged and transformed into a collective, positive community contribution. A crazy 8's design sprint started off the process. While pooling all the generated ideas together, we grouped their shared traits through affinity mapping. These themes informed how potential functions of the platform should (i) designate challenges which motivated individual actions on the part of the user, (ii) educate them on new actions that they could take to promote positive change and (iii) provide them with an innovative way of visualising the impact that both they and others are making as a collective. We also considered it important for potential users to anonymously share their impact on the community due to the reluctance to discuss contributions encountered in our research. For creating a mechanism of visualising impact and educating users, we decided to explore the area of augmented reality (AR) as it was a consistent theme in our crazy 8's sketches for its ability to make abstract concepts more tangible in the physical world.

4.1 Adoption of Augmented Reality Technology

Using augmented reality to stimulate awareness and engagement regarding social causes with a focus on the environment is a strategy that has been employed in recent years to great success. In a recent study, AR visualisations of food waste were reported to help university students understand the severity of food waste issues, with the majority reporting that it would make them more likely to waste less in the future [11].

Previous research on AR technologies' educational application stressed that the experience should not just be informative and easy to use but also fun and engaging [12]. As such, we came up with two novel design ideas that could help users share their contributions to social causes through AR with these goals in mind. While both ideas emphasise the element of communal effort and sharing contributions to raise awareness about social causes, they differ in terms of the scope of issues they address.

4.2 Idea I: Social Sphere

The social sphere is a public installation on which users can anonymously share their contributions to various social causes and inspire others. It was created from the feedback that individuals are motivated by seeing others' contributions but don't feel comfortable

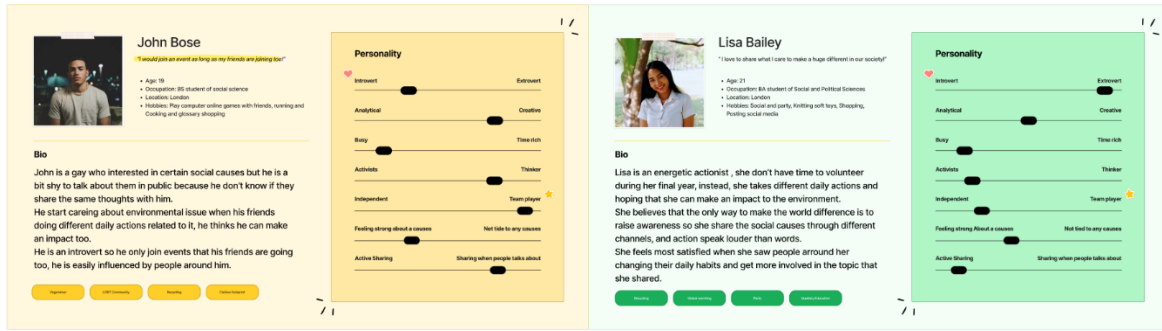


Figure 1: Primary (left) and secondary (right) personas

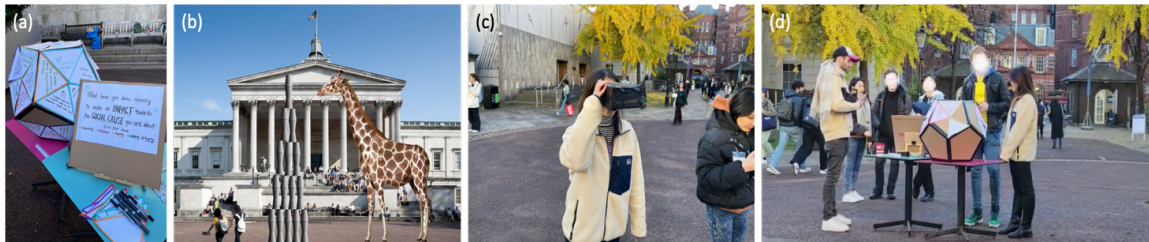


Figure 2: Testing initial ideas in the wild

sharing their own online. As individuals log their recent actions, these are represented through an AR visualisation of a sphere made up of tiles with pictures and corresponding descriptions of users' contributions.

4.3 Idea II: Coffee Cup Tower

The Coffee Cup Tower is also a public installation in a communal space informing students of the number of disposable coffee cups that can be saved as a collective. During an observation session in one of the college's cafes over the course of an hour, we found that only 12% of students used a reusable cup, despite a 25p discount being offered and reusable cups being sold at the venue. To solve this problem, the platform tracks the use of reusable coffee cups from the campus' cafes and visualises the number of disposable coffee cups that have been saved through AR.

4.4 Idea Evaluation

Prototypes for both concepts were displayed in the wild at the main campus entrance to mimic the public display aspect and gauge the level of interest and engagement with the potential user group (Figure 2d). For Social Sphere, a low-fidelity cardboard sphere prototype was created. People could add their recent contributions by drawing or writing on a piece of paper and placing it on a tile in the sphere (Figure 2a). Meanwhile, a cardboard headset with a phone at the end of it was utilised to evaluate the Coffee Cup Tower (Figure 2c), which displayed an image of a tower of coffee cups stacked on top of one another with familiar figures next to it like a giraffe and Big Ben (Figure 2b) to provide a comparable representation in the height of how many cups had been saved over a given period. The installation of Social Sphere garnered attention from more than

half of the passers-by. Some engaged both prototypes and gave feedback via short questionnaires.

4.4.1 Feedback for Social Sphere. The sphere was considered fun by 90% of the participants and inspired them to do new social actions that they hadn't done before. Many of the responses were regarding environmentally sustainable actions. In terms of displayed behaviour, some people were confident in writing down their contributions, while others were more reserved and claimed they were "just looking". These people may have had trouble thinking about what they had done recently on the spot or were more private about their personal behaviour and contributions.

4.4.2 Feedback for Coffee Cup Tower. The coffee cup tower was perceived to be a more intuitive, tangible experience in terms of showing impact. The visualisation of stacked cups gave a clear understanding of the scale of the issue and the communities' efforts to combat it. Participants considered the visualisation as educational, and 80% of them agreed that it would motivate them to use a reusable coffee cup and consider their environmental impact more. Participants also liked the idea of competing with other colleges partaking in the initiative to see who could build the highest tower on their own campus grounds.

5 CO₂LLAB

Considering that each concept had its own merits, we decided to incorporate elements of both into the final solution by including more contributions that people could make to help the environment. The contributions should also be tangible and concrete, while the AR visualisation should clearly reflect progress towards a goal for the community in an educational manner. Competition between

college communities in logging progress was another motivational factor for participating in the platform. With all this in mind, we shifted the scope of the platform to tracking multiple individual actions focused on reducing CO₂e emissions.

CO₂LLAB is a community-oriented platform that allows users to track and share their environmental contributions with others in their college through an app while discovering alternative ways that they can contribute towards reducing CO₂e emissions. Users log individual actions that are focused on reducing carbon emissions through a habit tracking app on their mobile phone. Examples of actions that they can log include eating a vegetarian meal, walking instead of vehicular transport, taking a quicker shower, using a reusable coffee cup and bringing your own shopping bag when shopping. These actions are then translated into CO₂e emissions savings. This savings impact is compared to alternative energy-draining activities such as phone charges or LED light bulbs to provide the user with more context as to their impact.

A big display screen placed at the front of campus forms the other half of CO₂LLAB's system (Figure 3). It shows users real-time updates of CO₂e emissions from all the buildings on the campus for a given month, which they are encouraged to try and offset with their logged actions. These CO₂e emissions will be visualised as bubbles in an AR visualisation linked to the public display (Figure 3), and it is the communities' responsibility to 'pop' these bubbles to clear the air. The kgs of CO₂e emissions offset as a collective will also be shown and grow bigger as individuals continue to log and add their contributions. The display screen also compared CO₂e emissions from different universities in a ranking system to incorporate a competitive element. Our hypothesis from our user research was that this would foster accountability within the community for their CO₂e emissions and motivate users to log their individual actions.

5.1 Similar Existing Applications

Numerous applications currently exist that allow users to track their environmentally friendly habits while measuring CO₂ emissions. One such application is Capture [13]. This allows users to input the meals they eat and modes of transport they use, calculating their CO₂ emissions over a given period. Similarly, there have been projects that allowed people to see the carbon footprint of their communities through visual representations. An example of this was a video made by the University of Plymouth, which visualised the scale of carbon emissions of the campus buildings through a pyramid of spheres stacked onto each other [14]. This was done to inform students of the scale of the issue and make it more relatable to them.

Our platform contains elements of both ideas, and to the best of our knowledge, nothing like this has ever been done in a similar setting utilising AR to foster community engagement to tackle an environmental issue like carbon emissions. As such, its novelty lies in its ability to bring people from the community together to take accountability for their actions, rather than being in isolation like with the Capture example, to view the scale of emissions in their community, similar to the University of Plymouth example, but distinct in that students try to offset these emissions through their individual actions as a collective.

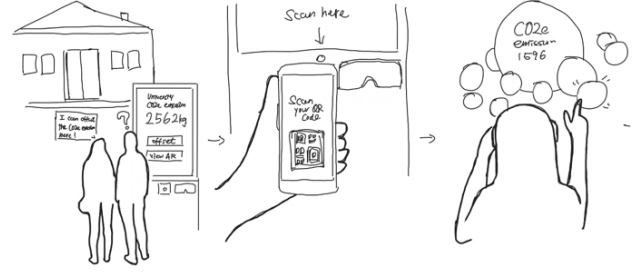


Figure 3: A sketch of the user flow of the public display & AR visualisation in CO₂LLAB

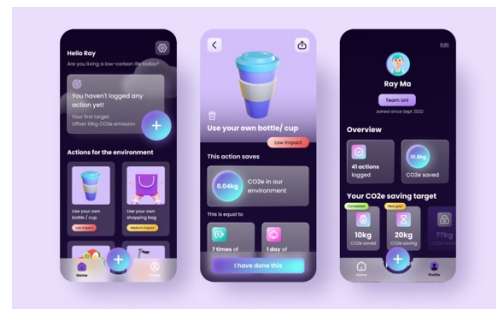


Figure 4: Mobile application interfaces

5.2 Prototype Evaluation

To evaluate the concept of CO₂LLAB, we tested a high-fidelity prototype with 4 target users using the think-aloud protocol. This included wireframes for the mobile application (Figure 4), a public display and a 3D model (Figure 5) for the CO₂e bubble-popping activity. Users were required to log actions on the interface, explore its various features and test the popping of CO₂e bubbles in a 3D model.

Overall, the tangible representations of contributions motivated users by showing the impact of their actions. Users liked seeing how others were contributing and enjoyed the university leaderboard feature. The public installation and mobile applications were perceived differently in terms of utilisation, with the former seen as more of a "campaign" that helps raise awareness of causes and what people do to contribute, while the latter accommodates the personal aspect of individual contributions and the convenience of sharing them whenever needed. Considering this, we removed the capability to log actions from the public display so that they can only be performed through the application.

It was also found that, as time passed, the proposed AR interaction could become repetitive. To overcome this, we extended the competitive aspect to the individual level by offering prizes that can be won after achieving a CO₂e savings milestone that will appear after a certain number of bubbles have been popped in the interaction. The prizes will be decided randomly as we want to ensure that reducing emissions remains the main focus for the users.

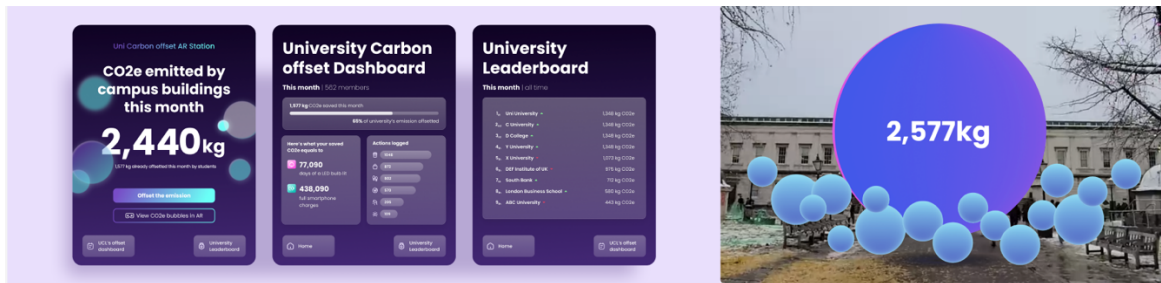


Figure 5: Public display interfaces (left) and 3D Model (right)

6 DISCUSSION

From our research, we discovered that young adults want to contribute to their cause but struggle with their fluid lives and busy schedules. As volunteering might not be feasible, they resort to individual actions, mainly focusing on creating environmentally sustainable habits. However, these individual actions can be easily forgotten or lose priority to more pressing daily events. They also become siloed from each other and tend to lose motivation as there is no clear direction of where their contribution is going. We viewed this as an opportunity to reimagine the way that young adults contribute to an environmental cause by encouraging them to offset their communities' CO₂e emissions through CO₂LLAB.

Sharing contributions within the community could significantly expand the positive impact that students otherwise doing individual actions alone can have on the environment. CO₂LLAB brings universities as the physical communities closer together to solve social issues through a novel implementation of AR technology. By seeing these visualisations physically, users can perceive environmental issues as more of a reality. We believe that CO₂LLAB could be an effective solution to goals 11 and 13 of the UN SDGs.

However, there are several challenges to CO₂LLAB. As it is in an early stage of development, the individual actions listed are still limited, and the social aspect is less pronounced in the mobile application. We only tested the visualisation as a 3D model, so the understanding of how effective the AR interaction would be is limited. Meanwhile, collaboration with relevant stakeholders would be essential to work out the real-time data visualisation of CO₂e emissions and the reward system. We incorporated feedback from prototype evaluation through several enhancements, such as creating individual targets through the reward system, but more tests are needed to ensure CO₂LLAB can better empower users to reduce CO₂e emissions.

7 CONCLUSION AND FUTURE WORK

CO₂LLAB was designed as a new mechanism for communities to tackle environmental issues together by tracking their individual habits to help them to understand the impact that their habits have on the environment and provide alternative ways that they can reduce this harm with more positive habits. In combination with this, its AR tool can provide a playful and educational mechanism for encouraging people to come together to offset the emissions they make as a community.

While it is a concept at the present time, it would be interesting to see such a platform rolled out across communities to gauge its effectiveness in changing individuals' behaviours to be more environmentally friendly. CO₂LLAB can also be extended to communities beyond universities if successful.

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